Neural Rosetta Stone: A multilingual model for Egyptian Hieroglyph Restoration

Elbaraa Hatem
Computer Science and Engineering
Alamein International University(AIU)
Egypt
Albara.ismail.2023@aiu.edu.eg

Amar Hassan
Computer Science and Engineering
Alamein International University(AIU)
Egypt
ammar.mohamed.2023@aiu.edu.eg

Abdelrahman Harb
Computer Science and Engineering
Alamein International University(AIU)
Egypt
abdelraheman.shehata.2023@aiu.edu.eg

Farah Mohamed
Computer Science and Engineering
Alamein International University(AIU)
Egypt
farah.khamis.2023@aiu.edu.eg

Karim Mahfouz
Computer Science and Engineering
Alamein International University(AIU)
Egypt
kmahfouz@aiu.edu.eg

Christine Basta
Computer Science and Engineering
Alamein International University(AIU)
Egypt
christine.raouf.saad.basta@upc.edu

***Abstract*—Centuries have weathered the stone of ancient Egypt, erasing words once carved to honor gods and kings. Traditional computer vision models attempt to resurrect these lost hieroglyphs by treating them as mere shapes—but what happens when the glyphs are too faint or entirely gone? In our work, we shift the paradigm: instead of seeing hieroglyph recovery as an image classification problem, we reframe it as a language problem—specifically, next-word prediction. Drawing from the grammatical and contextual richness of Egyptian inscriptions, we harness the strength of LSTM-based language models to infer what the eye cannot see. Our model, trained on scarce and static corpora, achieves over 48% accuracy—even with little context or in multi-word gaps—offering scholars a pragmatic, data-efficient tool to complete incomplete narratives. HieroLM doesn't just recognize signs—it remembers meaning.**

***Index Terms*—Recommender Systems, Popularity Bias, Reinforcement Learning, Long-tail Recommendations, Fairness, Diversity.**

I. INTRODUCTION

Egyptian hieroglyphs served as the formal writing system of Ancient Egypt, playing a central role in religious ceremonies and mortuary traditions. Decipherment requires a two-stage process: transliteration into phonetic representations followed by translation into modern languages [8], as exemplified in standard decoding demonstrations.

Archaeological artifacts frequently exhibit surface erosion, resulting in illegible or fragmentary hieroglyphs. While computer vision (CV) approaches have been developed for recognizing degraded symbols through image classification—typically using convolutional neural networks (CNNs) [2]–[4]—they face two critical constraints: label=()

1) **Visual dependency**: Severely damaged or missing glyphs cannot be processed due to exclusive reliance on visual features
2) **Contextual blindness**: Predictions disregard grammatical relationships and semantic context from adjacent words that could resolve ambiguities

For instance, consider an ambiguous symbol potentially representing either (*nb*) or (*sw*). Contextual analysis of a sentence describing a royal offering to Osiris would strongly favor (*sw*, "the king"). Similarly, entirely missing glyphs in a title position after Osiris' name could be inferred as (*dw*) based on the common epithet *nb dw* ("lord of Djedu") in offering formulae.

To address these limitations, we propose a language modeling approach that frames hieroglyph restoration as next-word prediction. Our architecture selection considers three linguistic characteristics of hieroglyphs [1]

1) **Finite corpora**: As a dead language, training data is inherently limited
2) **Domain specificity**: Usage was restricted to formulaic funerary, religious, and monumental contexts
3) **Local dependencies**: Sentences exhibit strong short-range patterns (e.g., titles following divine/royal names)

To capture these properties, we implement Bidirectional Long Short-Term Memory networks (BI-LSTMs) [9]. This architecture models contextual dependencies in both forward and backward directions—essential for reconstructing damaged or missing glyphs from surrounding linguistic context. we validate our design through comparative analysis with LSTMs [19] RNNs [13] and Transformers [18].

II. Related Work

A. *Hieroglyph Recognition with CV*

Modeling hieroglyph recognition as an image classification task is well-explored. [7] proposed using Histogram of Oriented Gradients (HOG) and Shape-Context (SC) descriptors for hieroglyph extraction and comparison. The HOG method was later enhanced with Region of Interest (ROI) extraction [6]. [14] and [2] evaluated ShuffleNet, MobileNet, ResNet, VGG, DenseNet, and Inception v3 architectures for hieroglyph recognition, while GlyphNet [4] achieves state-of-the-art performance. However, these computer vision models rely exclusively on visual features of individual signs and fail to incorporate linguistic context.

B. Next Word Prediction with LMs

Next-word prediction involves determining subsequent terms in sequences given preceding context. Early n-gram models suffered from data sparsity and limited context understanding. Neural Probabilistic Language Models (NPLM) [5] addressed these limitations using neural networks. Continuous Space Language Models (CSLM) [17] projected words into continuous vector spaces to handle variable-length contexts. Recurrent neural networks (RNNs) and Long Short-Term Memory (LSTM) networks [9] significantly improved prediction accuracy by maintaining memory states and capturing local dependencies. Recently, Transformers [18] revolutionized language modeling through self-attention mechanisms that consider entire input contexts, though their data requirements make them less suitable for hieroglyphic applications with limited corpora. For our work, we implement HieroLM [19] using Bidirectional LSTM (BI-LSTM) to leverage both preceding and subsequent context for damaged hieroglyph restoration.

III. Methodology

In this section, we describe in detail our BiLstm model, which adopts the bidirectional LSTM architecture for Egyptian hieroglyph sequence modeling.

Assume that the input sentence has T words. Let $x^{(t)} \in \{0,1\}^{|V|}$ be the one-hot encoding of the t-th word ($1 \leq t \leq T$) in the sentence. Then, its embedding $e^{(t)} \in \mathbb{R}^s$, where s is the embedding size, is computed as $e^{(t)} = Ex^{(t)}$, where E is an embedding layer.

Unlike the unidirectional LSTM, our BiLSTM processes the sequence in both forward and backward directions. The forward hidden state $\overrightarrow{h}^{(t)} \in \mathbb{R}^d$ and backward hidden state $\overleftarrow{h}^{(t)} \in \mathbb{R}^d$, where d is the hidden dimension size, at step t are computed as:

$$\overrightarrow{h}^{(t)} = F_{\theta_f}(\overrightarrow{h}^{(t-1)}, e^{(t)}) \tag{1}$$

$$\overleftarrow{h}^{(t)} = F_{\theta_b}(\overleftarrow{h}^{(t+1)}, e^{(t)}) \tag{2}$$

where F_{θ_f} and F_{θ_b} incorporate long short-term memory [9] for forward and backward directions respectively.

For the forward direction, given $\overrightarrow{h}^{(t-1)}$ and $e^{(t)}$, we compute the following states with single layer neural networks:

$$\overrightarrow{f}^{(t)} = NN^f_{forget}(\overrightarrow{h}^{(t-1)}, e^{(t)}) \tag{3}$$

$$\overrightarrow{i}^{(t)} = NN^f_{in}(\overrightarrow{h}^{(t-1)}, e^{(t)}) \tag{4}$$

$$\overrightarrow{g}^{(t)} = NN^f_{gate}(\overrightarrow{h}^{(t-1)}, e^{(t)}) \tag{5}$$

$$\overrightarrow{o}^{(t)} = NN^f_{out}(\overrightarrow{h}^{(t-1)}, e^{(t)}) \tag{6}$$

The forward cell state $\overrightarrow{c}^{(t)} \in \mathbb{R}^d$ at step t is computed as:

$$\overrightarrow{c}^{(t)} = \overrightarrow{f}^{(t)} \odot \overrightarrow{c}^{(t-1)} + \overrightarrow{i}^{(t)} \odot \overrightarrow{g}^{(t)}$$

where $\overrightarrow{c}^{(0)}$ is the initial forward cell state. The forward hidden state $\overrightarrow{h}^{(t)}$ is calculated as:

$$\overrightarrow{h}^{(t)} = \overrightarrow{o}^{(t)} \odot \tanh(\overrightarrow{c}^{(t)})$$

Similarly, for the backward direction, given $\overleftarrow{h}^{(t+1)}$ and $e^{(t)}$, we compute:

$$\overleftarrow{f}^{(t)} = NN^b_{forget}(\overleftarrow{h}^{(t+1)}, e^{(t)}) \tag{7}$$

$$\overleftarrow{i}^{(t)} = NN^b_{in}(\overleftarrow{h}^{(t+1)}, e^{(t)}) \tag{8}$$

$$\overleftarrow{g}^{(t)} = NN^b_{gate}(\overleftarrow{h}^{(t+1)}, e^{(t)}) \tag{9}$$

$$\overleftarrow{o}^{(t)} = NN^b_{out}(\overleftarrow{h}^{(t+1)}, e^{(t)}) \tag{10}$$

The backward cell state $\overleftarrow{c}^{(t)} \in \mathbb{R}^d$ and hidden state $\overleftarrow{h}^{(t)}$ are computed analogously:

$$\overleftarrow{c}^{(t)} = \overleftarrow{f}^{(t)} \odot \overleftarrow{c}^{(t+1)} + \overleftarrow{i}^{(t)} \odot \overleftarrow{g}^{(t)} \tag{11}$$

$$\overleftarrow{h}^{(t)} = \overleftarrow{o}^{(t)} \odot \tanh(\overleftarrow{c}^{(t)}) \tag{12}$$

where $\overleftarrow{c}^{(T+1)}$ is the initial backward cell state.

The bidirectional hidden representation at each time step t is obtained by concatenating the forward and backward hidden states:

$$h^{(t)} = [\overrightarrow{h}^{(t)}; \overleftarrow{h}^{(t)}] \in \mathbb{R}^{2d}$$

The predicted output is calculated by:

$$\hat{y} = NN_{pred}(h^{(T)})$$

where NN_{pred} is a single neural layer plus a softmax layer, which projects the final bidirectional hidden state from $2d$ to the size of the vocabulary $|V|$.

This bidirectional architecture allows the model to capture both past and future context information when predicting hieroglyphic sequences, potentially improving the understanding of complex hieroglyphic patterns and dependencies that exist in both directions within Egyptian texts.

IV. Experiments

A. Datasets

We evaluate our model and the baselines on four real-world datasets with hieroglyphic sentences from unearthed Egyptian artifacts. The dataset statistics are summarized in Table I.

- **Ancient Egyptian Sentences (AES)** [10]: A collection of over 100,000 ancient Egyptian sentences across multiple dynasties.
- **Ramses Transliteration Corpus** [16]: Contains transliterations converted from a large corpus of Late Egyptian sentences.
- **TLA Earlier_Egyptian** [20]: Contains 12,773 sentences with a vocabulary size of 3,000 tokens.
- **Mixed**: Since AES contains sentences from different eras while texts in Ramses come from Late Egypt, they

have different distributions due to language evolution. To evaluate the models' cross-distribution modeling ability, we synthesize AES and Ramses into a mixed dataset.
- **Combined**: We train our model on a combined dataset containing all available data.

We use the MdC transliterations of the hieroglyphic sentences throughout our experiments because it replaces irregular letters (e.g.,) and)') in common transliteration with English letters (e.g., "a" and "A") for convenient processing. The sentences are split into training, validation, and test sets by an 8:1:1 ratio for individual datasets, while the combined dataset uses custom splits.

TABLE I
DATASET STATISTICS

Dataset	Sentence #	Training #	Validation #	Test #
AES	98,375	78,801	9,800	9,774
Ramses	61,069	48,848	6,116	6,105
TLA	12,773	-	-	-
Mixed	159,444	127,649	15,916	15,879
Combined	318,928	255,298	31,832	31,798

B. Baselines

We evaluate our BiLSTM model against the following established baselines:

- Neural Probabilistic Language Model (NPLM) [5]. We implement a trigram NPLM as the simplest baseline.
- Recurrent Neural Network (RNN) [13]. We use a single-layer, unidirectional RNN, serving as a basic recurrent model.
- LSTM-based HieroLM [19]. We include the LSTM model from Cai and Zhang (2024), designed for Egyptian hieroglyph recovery with next word prediction.
- Transformer [18]. We utilize a single-layer encoder with 16 attention heads and zero dropout, constrained by limited data.

C. Performance Validation

We present the key findings in Table II, leading to the following insights:

- **Limited hieroglyphic vocabulary.** Predicting the next word is inherently challenging due to the flexibility of modern languages, where multiple valid words can grammatically and semantically fit a given context. For comparison, Google's state-of-the-art LSTM-based language model for English, trained on billions of words, achieves a perplexity of 30 [11]. In contrast, BILSTM records a perplexity of approximately 19 with under a million words, suggesting that the hieroglyphic vocabulary is notably restricted.
- **Superiority of recurrent architectures.** The table indicates that models with recurrent structures, such as BILSTM and LSTM, consistently outperform others on small datasets. This highlights the strength of recurrent models in capturing local semantic relationships even with limited data.
- **LSTM's performance boost.** Comparing BILSTM with RNN provides a direct ablation study. BILSTM's superior performance over RNN demonstrates that BILSTM's ability to capture long-range dependencies enhances the model's effectiveness.
- **Transformer's unsuitability for this task.** The results show that the Transformer lags behind BILSTM, indicating that its architecture is less effective for this task, likely due to the constraints of limited data.

TABLE II
KEY PERFORMANCE RESULTS ACROSS DATASETS.

Dataset	Metric	BILSTM	Transformer	RNN	HieroLM
AES	Perplexity	**24**	52.21	42.25	26.50
	Accuracy	**0.4981**	0.3143	0.3828	0.4525
	F1 Score	**0.2374**	0.0488	0.1201	0.1420
Ramses	Perplexity	**19**	38.59	31.89	21.59
	Accuracy	**0.5142**	0.3727	0.4387	0.4895
	F1 Score	**0.3457**	0.0905	0.1933	0.2074
Combined	Perplexity	**22.451**	53.78	43.34	26.48
	Accuracy	**0.4795**	0.3151	0.3801	0.4450
	F1 Score	**0.2134**	0.0466	0.1377	0.1421

D. Multi-shot Prediction Performance

In reality, it is common for a number of contiguous hieroglyphic words to be missing together, which makes it important to evaluate the model's ability to predict a series of words accurately without teacher forcing. Table 3 presents the accuracy of BiLSTM for multiple following words. We can observe a favorable diminishing decrease in accuracy with the increase of prediction range. It is also worth noting that the model maintains an accuracy of over 14% on predicting 4 words in a row.

TABLE III
MULTI-SHOT PREDICTION ACCURACY.

Dataset	1 Shot	2 Shot	3 Shot	4 Shot
AES				
LSTM	0.45	0.20	0.16	0.14
BiLSTM	0.50	0.25	0.18	0.15
Ramses				
LSTM	0.48	0.28	0.20	0.17
BiLSTM	0.52	0.30	0.24	0.19

E. Resistance against Data Scarcity

A significant challenge in applying machine learning techniques to hieroglyphic reconstruction is the limited availability of data, which presents difficulties at two distinct levels: At the corpus level, the overall quantity of hieroglyphic sentences derived from ancient artifacts remains constrained. At the sentence level, numerous hieroglyphic texts are fragmentary due to deterioration over time, containing only a small number of recognizable symbols. This limited contextual information compounds the challenge of deducing missing words.

V. Conclusion

In this study, we frame Egyptian hieroglyphic reconstruction as a sequential word prediction problem tackled through language modeling approaches. Given the dataset constraints and localized semantic relationships, we introduce BiLSTM with recurrent neural network architecture, which demonstrates substantial accuracy in our experimental evaluation. The model's remarkable performance in multi-step predictions and its effectiveness with limited input sequences render it valuable for archaeological applications in reconstructing missing hieroglyphs and enhancing computer vision approaches. Moving forward, we intend to investigate methodologies for combining computer vision and language modeling frameworks into a comprehensive and robust hieroglyphic restoration system.

VI. Limitations

In this study, the constrained dataset size prevented us from effectively harnessing the capabilities of contemporary Transformer architectures. Although adapting Transformer models to smaller datasets is feasible, such adaptation necessitates advanced training methodologies [15] and has been shown to exhibit reduced stability in certain scenarios [12]. Future research directions will focus on investigating methods to customize self-attention mechanisms for Egyptian hieroglyphic corpora. Additionally, we plan to access the complete TLA (Thesaurus Linguae Aegyptiae) dataset to expand our training data and potentially enable more effective utilization of Transformer-based models.

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